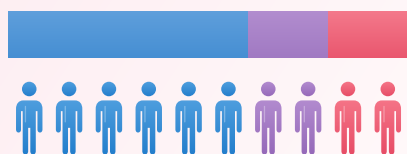


A

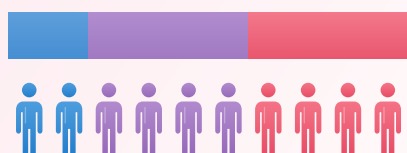
Per-group estimation: the estimand depends on the group's composition

1. Group composition (different across groups)

Mostly healthy
(e.g. ϵ_2 carriers)

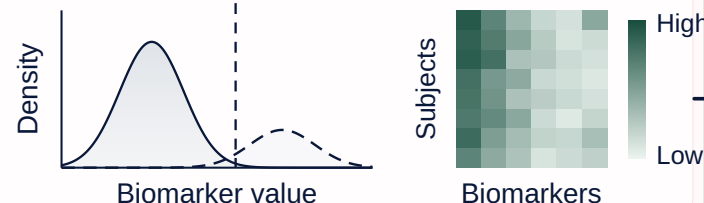


Mostly impaired
(e.g. ϵ_4 carriers)



2. Mixture model & subject event probabilities (fit separately in each group)

Mostly healthy
(few abnormal subjects: small weight)



Mostly impaired
(many abnormal subjects: large weight)



3. Consensus over subjects (equal weight within each group)

Mostly healthy
(more early-stage subjects)



Unweighted average
(equal weight per subject)

Mostly impaired
(more late-stage subjects)



Unweighted average
(equal weight per subject)

4. Estimated ordering (different targets)

True ordering (same in both groups)



Mostly healthy
(larger deviation, example)



Cognitive events (D, E) earlier
and imaging event (C) later
in healthy-heavy group (example)

Mostly impaired
(smaller deviation)



5. Permutation test (labels exchanged among all subjects)

Permuted group 1 Same composition
(in expectation)



Permuted group 2



Null distribution



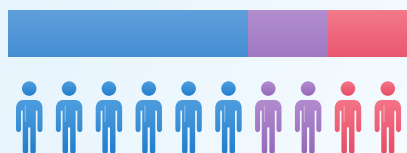
**Inflated
false positives**
due to different
targets

B

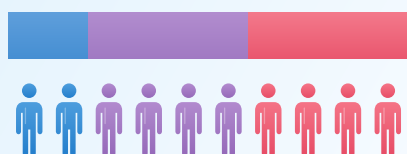
Composition-invariant estimation: one measurement model, one reference composition, stratified permutation

1. Group composition (same data as A)

Mostly healthy
(e.g. ϵ_2 carriers)

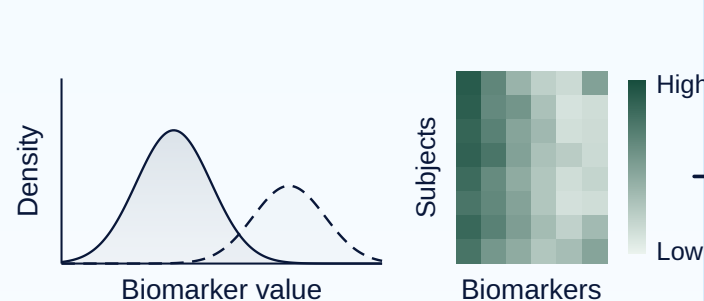


Mostly impaired
(e.g. ϵ_4 carriers)



2. Pooled mixture model & subject event probabilities (fit once on all subjects)

All subjects (pooled)



Fit on all subjects

3. Consensus over subjects (weighted to a common composition)

Mostly healthy
(down-weight CN, up-weight MCI/AD)



Weighted average
(to common composition)

Mostly impaired
(down-weight AD, up-weight CN)



Weighted average
(to common composition)

Common reference composition
(same in both groups)



4. Estimated ordering (same target)

True ordering (same in both groups)



Mostly healthy
($\approx$ true, noise only)



Mostly impaired
($\approx$ true, noise only)



Both groups estimate the same target
(differences are noise only)

5. Permutation test (labels exchanged within diagnosis strata)

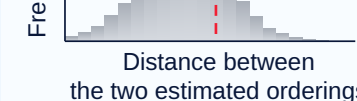
Permuted group 1



Permuted group 2



Null distribution



Valid inference
(composition
controlled)

